Module 3 — Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 3, Quiz A: Definitions & notation

Reading: Module 3 Read 1
Project: Project 3 · Canvas: Def & Notation quiz (Module 3)

Work these practice items after Read 1 (Sections 1–3: one categorical, one numerical, numerical × categorical). Sections cover: variable types, hand computation (counts, proportions, IQR, five-number summary), plot identification, and graph image items. (R/tidyverse questions live in the Synthesis quiz.)

Graph application, scatter interpretation, and two-way tables → Quiz B after Read 2.

Section A — Variable types and one-variable summaries

A1. Numerical vs categorical

Classify each as numerical or categorical (and binary if categorical):

a) Penguin flipper length (mm)

b) Island: Biscoe / Dream / Torgersen

c) Sex: male/female

d) Clutch completion: yes/no

e) Bill depth (mm)

A2. Best summary for one numerical variable

For one numerical variable, name one graph and two numeric summaries.

A3. Best summary for one categorical variable

For one categorical variable, name one graph and one numeric summary.

A4. Choosing median or mean by purpose

Your dataset has a few very large values. For which kind of question would you report the median, and for which the mean?

Answer key — Section A — Variable types and one-variable summaries

1. a) Numerical. b) Categorical, not binary. c) Categorical, binary. d) Categorical, binary. e) Numerical.
2. Graph: histogram (or dot plot/boxplot). Numeric: a measure of center and a measure of spread — median + IQR (or MAD) to describe a typical value and its spread, or mean + SD when you want the mean and its companion measure (e.g. heading toward a confidence interval).
3. Graph: bar chart. Numeric: counts or proportions per category.
4. It depends on the purpose. Report the median when the question is about a typical member's experience — the middle value, half above and half below — since it is not moved by a few extreme values. Report the mean when the question is about a total shared across units (total ÷ n), such as a budget or per-unit amount; you want the mean even when the data are skewed, because the total is what matters.

Section B — Spread: range, IQR, and standard deviation

B1. Range

Data: $4, 6, 8, 10, 10, 12, 14, 16$. Compute the range.

B2. Quartiles and IQR

Dataset N: $7, 8, 9, 10, 10, 11, 12, 13$. Find Q_1, Q_3, and IQR.

B3. Five-number summary

List the five numbers in the five-number summary in order.

B4. What a boxplot shows

Describe how a boxplot represents the five-number summary.

B5. SD interpretation

In words, what does the sample standard deviation s measure?

B6. IQR vs SD

When would you prefer IQR over SD to describe spread? Why?

Answer key — Section B — Spread: range, IQR, and standard deviation

1. Range $= \max - \min = 16 - 4 = 12$.
2. Lower half: $7, 8, 9, 10 \to Q_1 = (8+9)/2 = 8.5$. Upper half: $10, 11, 12, 13 \to Q_3 = (11+12)/2 = 11.5$. IQR $= 3$.
3. Minimum, Q_1 (first quartile), Median (Q_2), Q_3 (third quartile), Maximum.
4. The box spans Q_1 to Q_3; the line inside is the median; whiskers extend to the min and max (or fences if outliers are shown as dots).
5. s measures the typical distance of individual observations from their mean — a larger s means values are more spread out from $\bar{x}$.
6. When the goal is simply to describe the spread of one variable: the IQR (the width of the middle 50%) is a span you can read directly in the data's own units, with no normality assumption, and the MAD works similarly. SD is mainly a building block for inference — it feeds the standard error and confidence intervals — and is most directly interpretable when the distribution is roughly normal (empirical rule); it is also more sensitive to extreme values than the IQR.

Section C — Hand computation: counts and proportions

C1. Coffee order counts

Coffee shop ($n = 12$ orders): Latte 5, Drip 4, Tea 3. How many Latte orders? What is $\hat{p}$ for Latte?

C2. Survey yes/no

Survey ($n = 20$): Yes 13, No 7. Compute $\hat{p}$ for Yes.

C3. Bird feeder species

Feeder ($n = 30$ visits): Robin 12, Sparrow 11, Other 7. What proportion are Robin?

Answer key — Section C — Hand computation: counts and proportions

1. Count $x = 5$. $\hat{p} = 5/12 \approx 0.417$ (about 41.7%).
2. $\hat{p} = 13/20 = 0.65$ (65%).
3. $12/30 = 0.40$ (40%).

Section D — Compute each statistic by hand (one list: 7, 8, 9, 10, 10, 11, 12, 13)

D1. Mean by hand

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. Compute the mean $\bar{x}$.

D2. Median by hand

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. Compute the median.

D3. Minimum

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. What is the minimum?

D4. Maximum

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. What is the maximum?

D5. First quartile Q_1

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. Compute Q_1.

D6. Third quartile Q_3

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$. Compute Q_3.

D7. IQR by hand

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$ (with $Q_1 = 8.5$, $Q_3 = 11.5$). Compute the IQR.

D8. Sample SD by hand

Data ($n = 8$): $7, 8, 9, 10, 10, 11, 12, 13$ (mean $= 10$). Compute the sample standard deviation s.

Answer key — Section D — Compute each statistic by hand (one list: 7, 8, 9, 10, 10, 11, 12, 13)

1. $\bar{x} = (7+8+9+10+10+11+12+13)/8 = 80/8 = 10$.
2. With $n = 8$ (even), median = average of the 4th and 5th values = $(10+10)/2 = 10$.
3. Minimum = 7 (the smallest value).
4. Maximum = 13 (the largest value).
5. Lower half: $7, 8, 9, 10$. $Q_1 = (8+9)/2 = 8.5$.
6. Upper half: $10, 11, 12, 13$. $Q_3 = (11+12)/2 = 11.5$.
7. IQR $= Q_3 - Q_1 = 11.5 - 8.5 = 3$.
8. Squared deviations from 10: $9, 4, 1, 0, 0, 1, 4, 9$, summing to 28. $s^2 = 28/(8-1) = 4$; $s = \sqrt{4} = 2$.

Section E — Plot types: bar, pie, histogram, dot plot, boxplot (verbal)

E1. Bar chart purpose

You have species (Adelie / Chinstrap / Gentoo). Which graph shows counts per level?

E2. Pie chart vs bar

When is a pie chart acceptable for one categorical variable?

E3. Pie slice interpretation

A pie chart shows 30% Latte, 25% Drip, 45% Tea ($n = 200$ orders). How many Tea orders?

E4. Histogram vs bar

Histogram bins a numerical variable; bar chart has one bar per category. Which fits flipper length (mm)?

E5. Dot plot use

When is a dot plot especially helpful for one numerical variable?

E6. Boxplot five-number summary

A boxplot shows min=5, $Q_1 = 10$, median=12, $Q_3 = 20$, max=50. Is the distribution skewed? Which way?

E7. Side-by-side boxplots

You compare body mass across three species. Name the graph.

E8. Histogram skew (verbal)

A histogram has a long right tail with most values piled on the left. Describe shape.

E9. Dot plot vs histogram

You have one numerical variable with $n = 15$ observations and want to see every individual value. Dot plot or histogram?

Answer key — Section E — Plot types: bar, pie, histogram, dot plot, boxplot (verbal)

1. Bar chart — one bar per category.
2. When there are few categories and you want to show parts of a whole as proportions — but a bar chart is usually easier to read.
3. $0.45 \times 200 = 90$ Tea orders.
4. Histogram (numerical variable).
5. With a small or moderate sample where you want to see each value stacked along the axis.
6. Right-skewed — long upper whisker; median closer to Q_1.
7. Side-by-side boxplots (numerical $\times$ categorical).
8. Right-skewed (positively skewed).
9. Dot plot — one dot per value (stacked).

Section F — Statistics: interpret counts, proportions, center, spread (3 each)

F1. Count interpretation

In a frequency table of species, the count $n = 124$ for Gentoo. What does 124 mean?

F2. Total from counts

Counts: A=8, B=12, C=5. What is n?

F3. Proportion from count

$x = 9$ successes out of $n = 36$. State $\hat{p}$.

F4. Mean vs median choice

Income data with a few very high earners. You want to describe a typical household. Report center with mean or median?

F5. SD interpretation 1

$s = 2$ cm vs $s = 15$ cm for the same variable. Which sample is more spread out?

F6. IQR interpretation

IQR = 4 vs IQR = 20 for two datasets. Which middle 50% is tighter?

F7. Range sensitivity

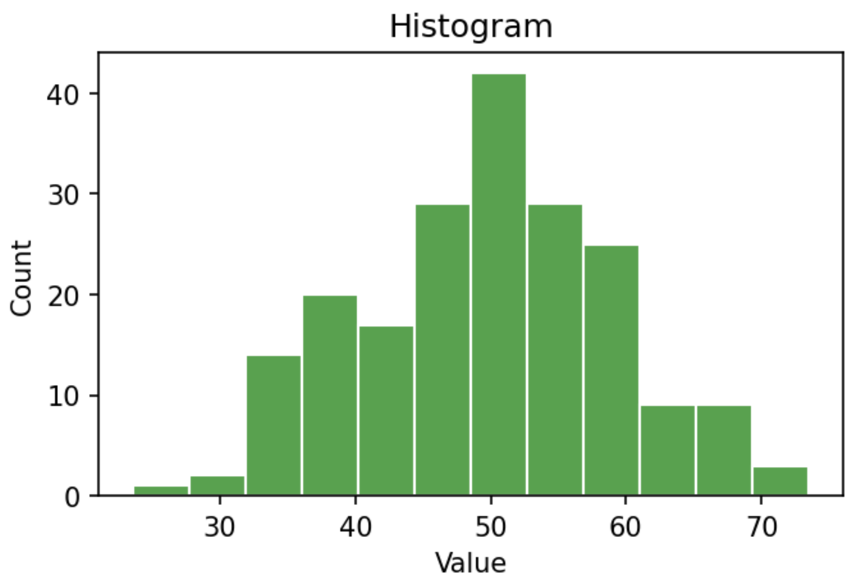
Why is range less useful than IQR when outliers exist?

Answer key — Section F — Statistics: interpret counts, proportions, center, spread (3 each)

1. 124 observations (penguins) in the sample have species = Gentoo.
2. $n = 8 + 12 + 5 = 25$.
3. $\hat{p} = 9/36 = 0.25$ (25%).
4. Median — the question is about a typical household (the middle value), which a few very high earners do not move. (If the question were instead about a total or per-capita amount, e.g. a budget, you would use the mean.)
5. $s = 15$ cm — larger typical distance from the mean.
6. IQR = 4 — middle half spans only 4 units.
7. Range uses only min and max — one extreme value can make range huge.

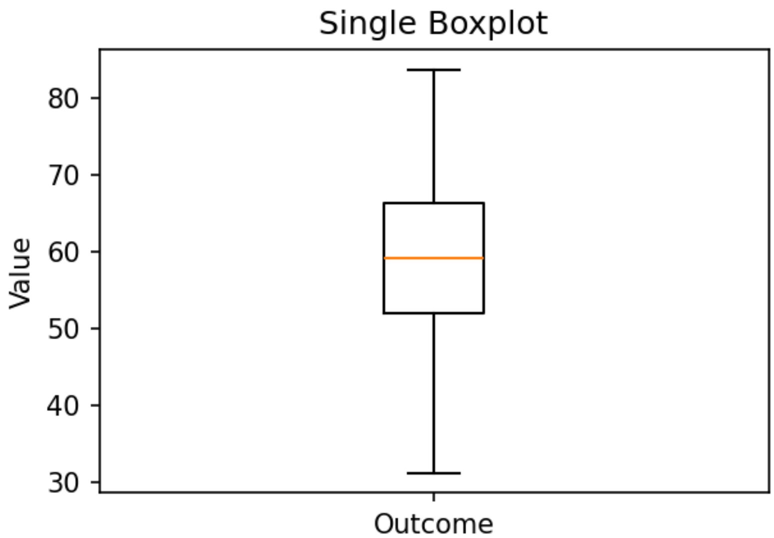
Section G — Graph interpretation (images · Quiz A)

IMG4. Histogram interpretation



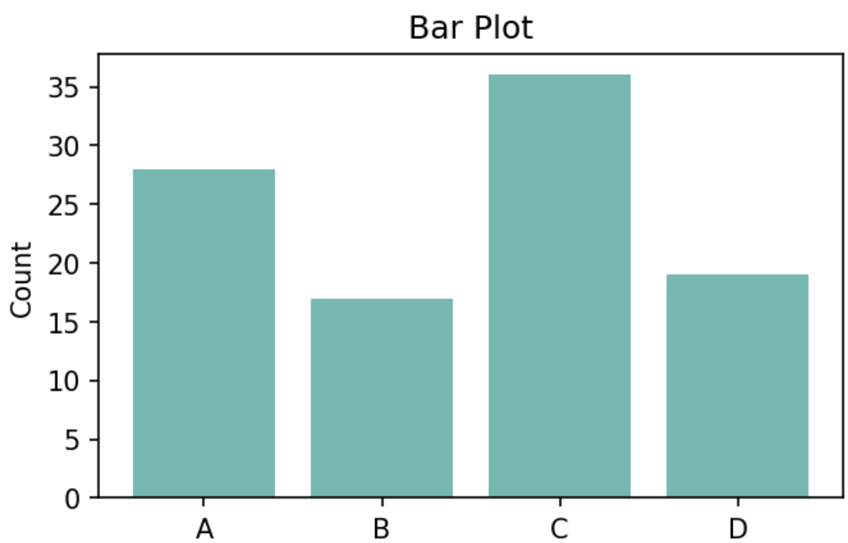
What does this graph primarily display?

IMG6. Single boxplot feature



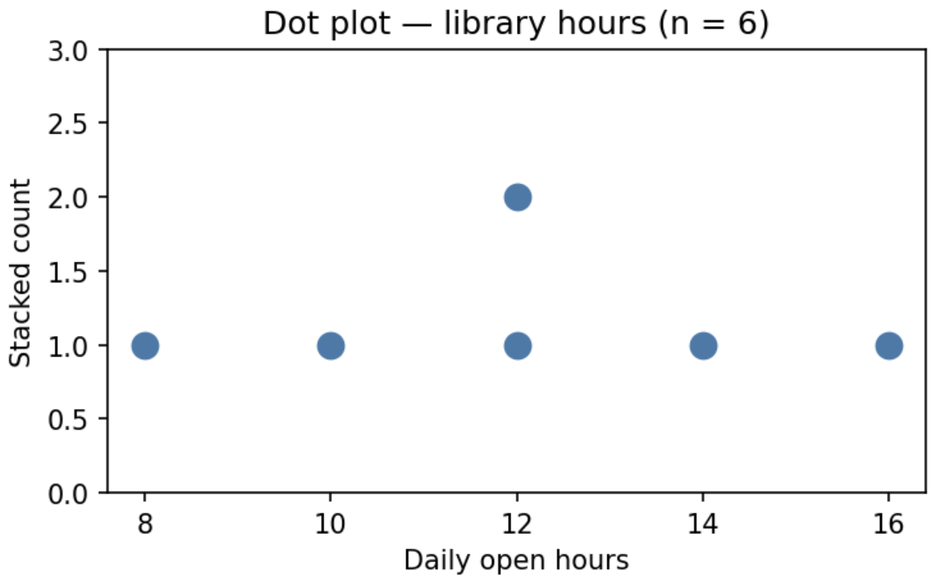
In a boxplot, the horizontal line inside the box marks the:

IMG9. Bar plot variable type



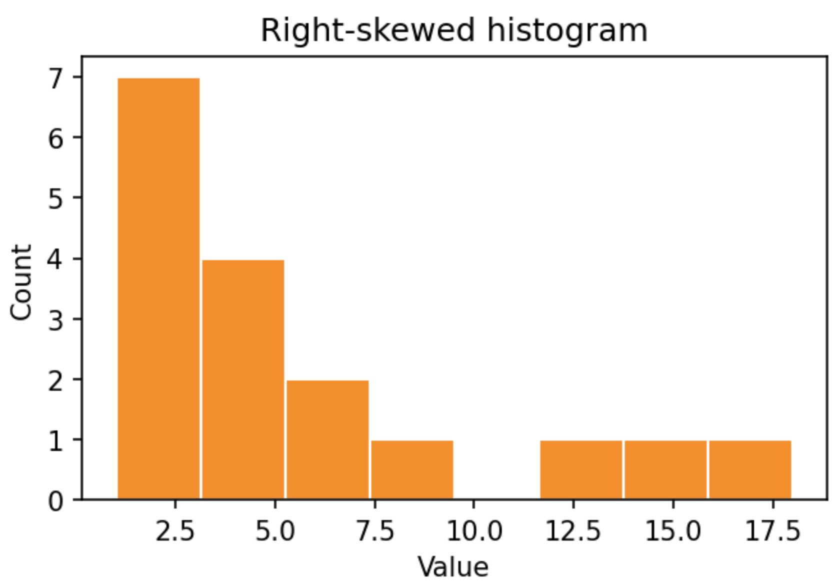
The x-axis categories in a bar plot represent what variable type?

IMG12. Dot plot interpretation



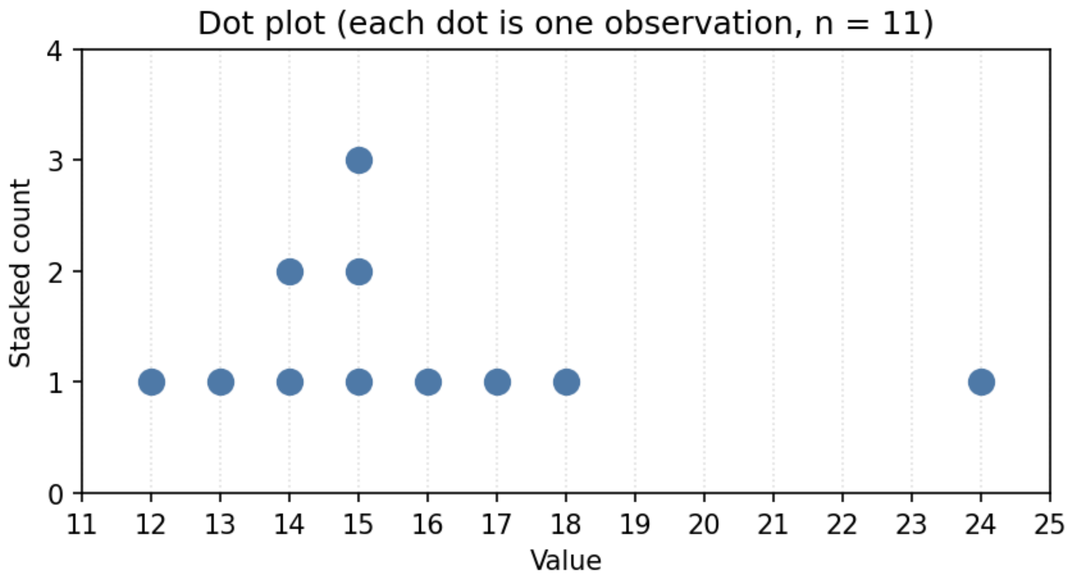
What does this dot plot primarily display?

IMG13. Right-skewed histogram



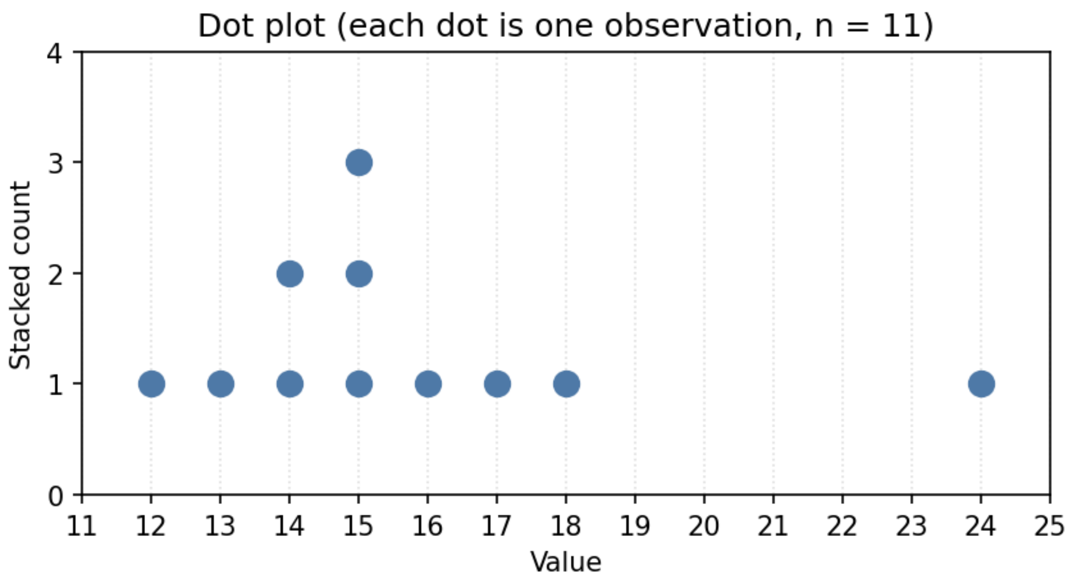
Describe the shape of this histogram.

IMG14. Dot plot — estimate the center



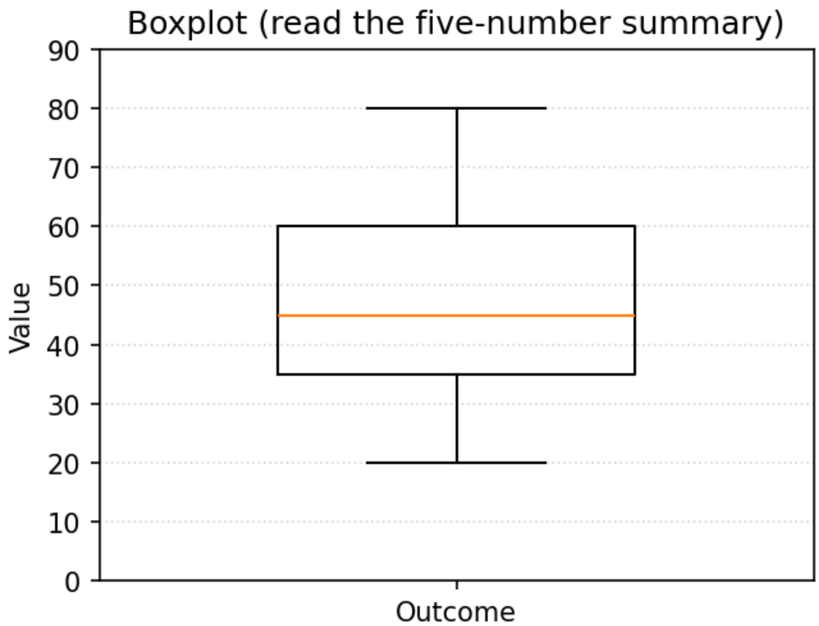
Each dot is one observation. Which value best estimates the center (median) of this dot plot?

IMG15. Dot plot — estimate the range



Estimate the range (largest value minus smallest value) shown in this dot plot.

IMG16. Boxplot — estimate the IQR



Read this boxplot. Estimate the IQR (the height of the box, Q3 − Q1).

Answer key — Section G — Graph interpretation (images · Quiz A)

1. The distribution of one numerical variable.
2. The median.
3. Categorical.
4. The distribution of one numerical variable — each dot is one observation (stacked).
5. Right-skewed (positively skewed) — long tail to the right; most values pile on the left.
6. About 15 — counting to the middle dot lands at 15, where most values cluster.
7. About 12 — values run from about 12 up to about 24, so range $24 - 12 = 12$.
8. About 25 — the box runs from about Q1 = 35 to about Q3 = 60, so IQR 25.
